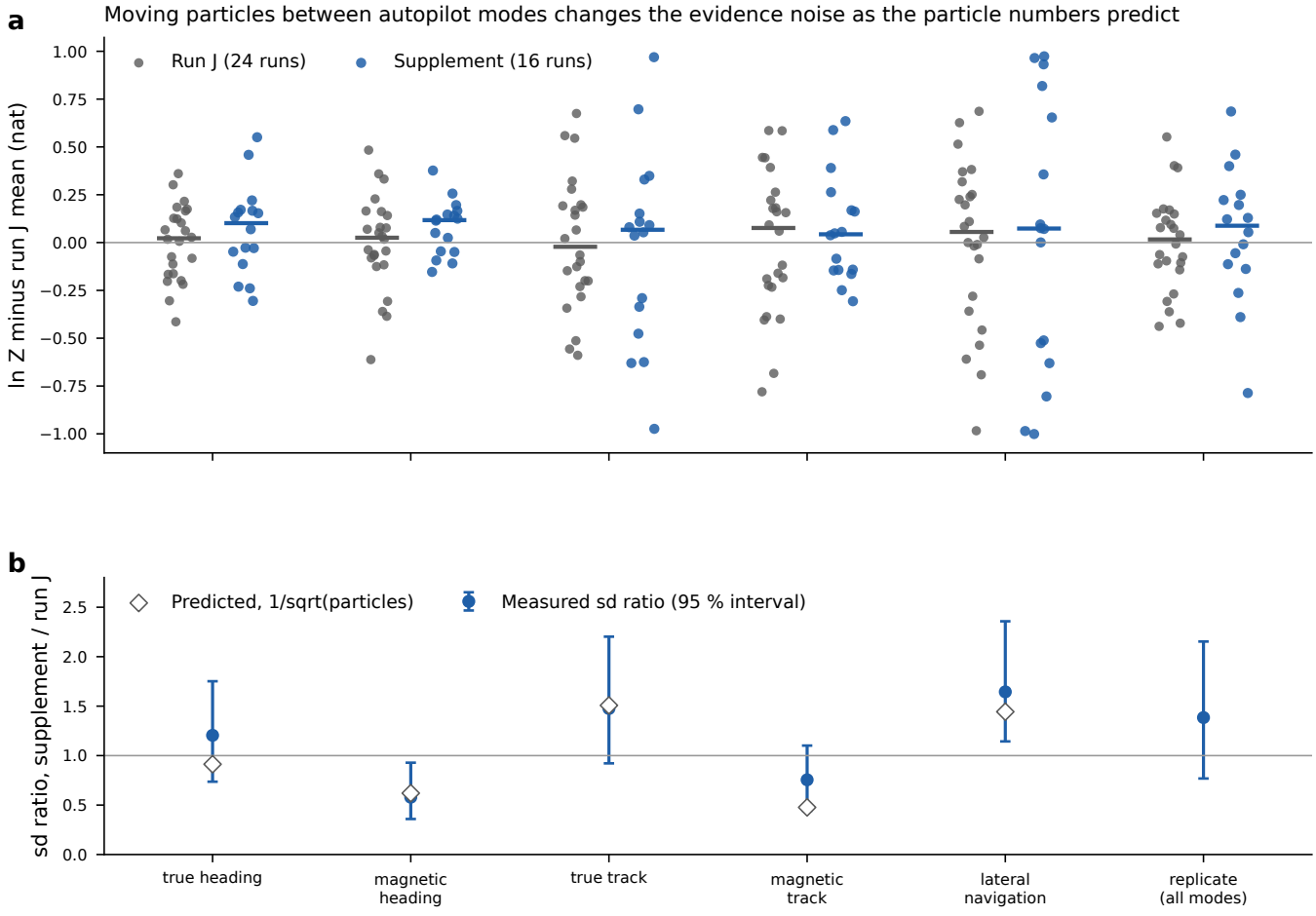

Plain: Panel a shows the evidence of each autopilot mode in each run, measured from the run J mean. The supplement runs have more particles in the magnetic modes and fewer in the true-track and lateral-navigation modes. Panel b compares the measured spread with the spread that the particle numbers predict. For most modes they agree, so the random error of the mode probabilities comes mainly from the particle number. The source is the core filter: run J, early-flight family 'waypoints' (24 runs), and the northern supplement (16 new runs). The prior track starts from the 18:01 radar position and follows a random route through the 2014 waypoints. The 00:11 data are used; the 00:19 data are not. The one-engine physics and the fuel-flow factor are provisional. No search areas, ocean drift or end-of-flight data are applied. The 18:25:27 signal is not used. The other three early-flight families are not included.

Technical: ln Z per mode = log evidence of each mode filter (diagnostics.json), run J waypoints seeds 1-24 against supplement seeds 1001-1016; 'replicate' = ln Z of the mode mixture. Panel b: ratio of seed sd (supplement / run J) with 95 % bootstrap interval (4,000 resamples of seeds), against sqrt(N_run J / N_supplement) for N particles per mode, the ratio expected if the sd falls as 1/sqrt(N). Mean ln Z per mode agrees with run J within 0.1 nat (target unchanged). Core 00:11 hand-offs (filtering distribution at m0011, look-ahead correction exp(log_correction) applied): run J waypoints seeds 1-24 (tag run-j, 1244b7c; 100k rows per seed) and supplement seeds 1001-1016 (rev 3bf0801; particles per mode [TH, MH, TT, MT, LN] = [1.2, 1.3, 1.1, 2.2, 1.2]M vs run J [1.0, 0.5, 2.5, 0.5, 2.5]M; supplement hand-off 100k rows per seed, tilt r(lat) = 0.1 + 0.9 h/H over 00:11 latitude -34.5..-21.0). BFO sd 4.3177 Hz; kappa_x ~ N(1.0, 0.046) and one-engine s7 + s8 PROVISIONAL; 18:25:27 R600 held out. Search case: none; ocean models: none; end-of-flight 00:19 options: not applied. Combination rule B: balance heuristic with r_hat (r_hat.csv), n1 = 2,399,999, n2 = 1,600,003 rows.